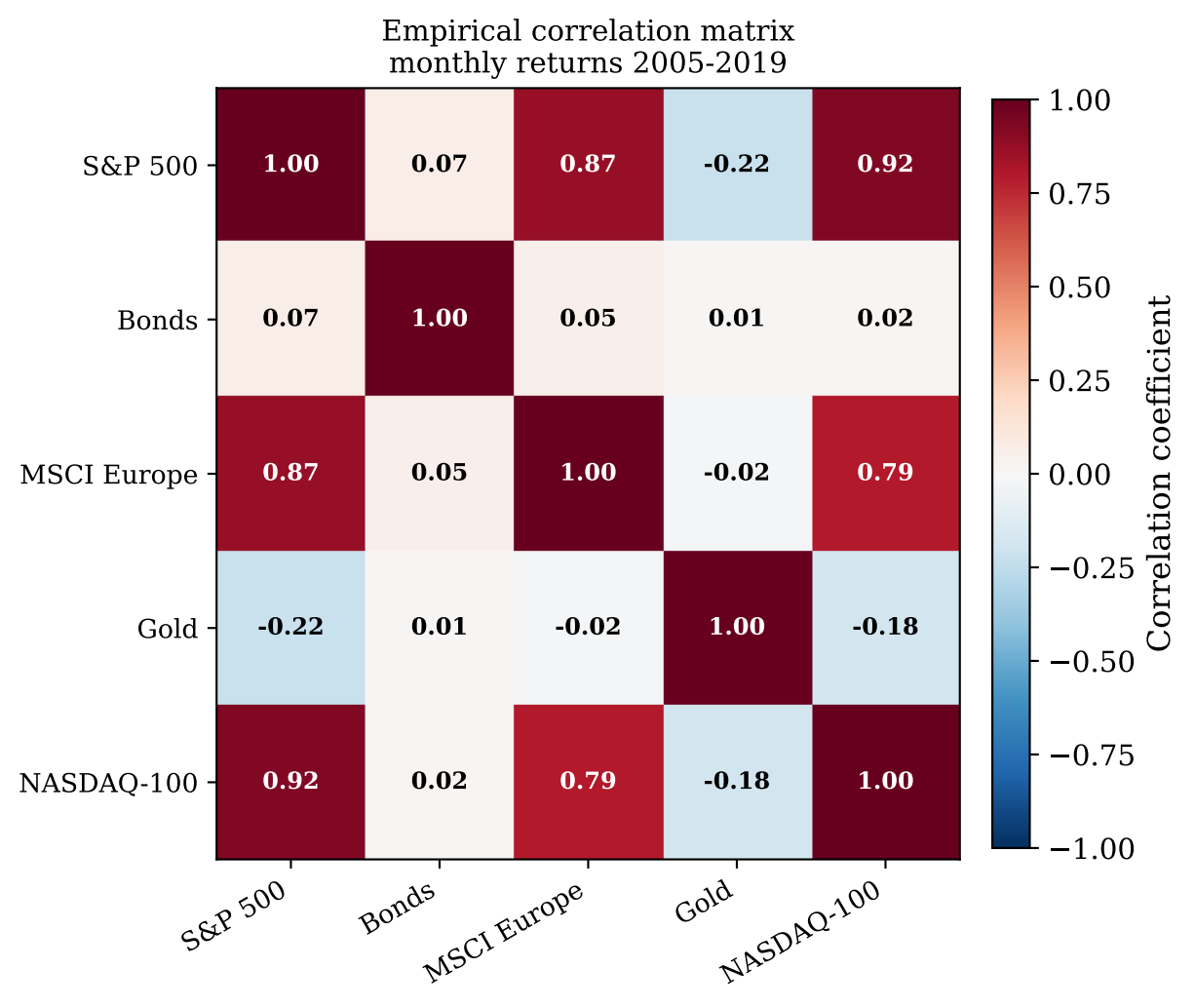
Empirical correlation matrix
monthly returns 2005-2019
S&P 500
Bonds
MSCI Europe
Gold
NASDAQ-100
1.00 0.07 0.87 -0.22 0.92
0.07 1.00 0.05 0.01 0.02
0.87 0.05 1.00 -0.02 0.79
-0.22 0.01 -0.02 1.00 -0.18
0.92 0.02 0.79 -0.18 1.00
S&P 500
Bonds
MSCI Europe
Gold
NASDAQ-100
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Correlation coefficient